

Amjad Alolayan

GIS and Spatial Analysis Studies | Sustainable Urban Development

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Portfolio: joodmicmuffin-bit.github.io | **GitHub:** github.com/joodmicmuffin-bit



SUMMARY

Master's student in Sustainable Urban Development at TU Darmstadt (expected graduation 2027), specializing in GIS and spatial analysis. Experienced in ArcGIS Pro, Python geospatial workflows (GeoPandas, Shapely, STRtree), and multi-criteria suitability modeling for renewable energy siting. Certified in ArcGIS API for Python and ArcGIS Pro scripting and spatial automation. Background in compliance, environmental regulation, and policy research. Actively seeking Werkstudent and Praktikum opportunities in GIS, renewable energy, and urban planning across Germany. Open to relocation.

WORK EXPERIENCE

Compliance and AML Officer | Gulf International Bank

Al Khobar, Saudi Arabia | April 2022 - September 2023

- Reviewed customer transactions and account activity to identify and report anti-money laundering risks in line with regulatory requirements.
- Conducted policy and risk analysis supporting regulatory compliance.
- Prepared compliance documentation and supported audits by maintaining accurate records of due diligence checks.

Intern | National Centre for Environmental Compliance (NCEC)

Riyadh, Saudi Arabia | May 2021 - October 2021

- Supported environmental compliance and inspection projects, monitoring facility adherence to environmental regulations through data collection, policy documentation, and preparation of governmental reports.

Research Intern | JACES (Institute for Global Environmental Strategies)

Tokyo, Japan | July 2017 - September 2017

- Conducted policy research and contributed to recommendations under the SDGs/Sustainable Consumption and Production (SCP) Program, including literature review, data analysis, and English-Japanese translation of policy documents.

GIS PROJECTS | github.com/joodmicmuffin-bit

Solar Park Site Suitability Analysis | Wetteraukreis and Vogelsbergkreis, Hesse

Personal Project | 2026

- Multi-criteria suitability analysis in ArcGIS Pro and Python (GeoPandas, Shapely, STRtree) across two contrasting districts; screened 31,699 parcels down to 197 buildable sites ranked on five weighted criteria (EEG transport corridor, substation distance, slope, aspect, solar irradiation), published as an interactive Folium and Leaflet web map.

City Population Share Analysis | Federal States of Germany

Personal Project | 2026

- Built a reusable ArcGIS Pro ModelBuilder tool automating each city's share of its federal state's total population across four contrasting states, using spatial joins, dissolves, and parameterized field calculations in Python.

Solar Energy Potential on Roof Surfaces | Breuberg, Hesse

Academic Coursework, independently rebuilt | 2026

- Estimated rooftop solar PV potential for 3,809 buildings using ArcGIS Pro, OpenStreetMap building footprints, and PVGIS annual irradiation data, correcting the original coursework's monthly-radiation error.

EDUCATION

M.Sc. Sustainable Urban Development | TU Darmstadt

Darmstadt, Germany | October 2023 - Present (Expected 2027)

Focus on GIS, spatial analysis, and sustainable planning; thesis planned on post-occupancy evaluation in green buildings.

Joint Master's Program | Vietnamese German University

Ho Chi Minh City, Vietnam | October 2024 - March 2025

Exchange semester within the dual-degree Sustainable Urban Development program.

B.A. Community and Regional Policy Studies | Ritsumeikan University

Kyoto, Japan | September 2016 - March 2021

Full scholarship; coursework in regional policy, sustainability, and environmental governance.

SKILLS

GIS and Mapping: ArcGIS Pro, QGIS, kernel density analysis, multi-criteria suitability modeling

Programming: Python, GeoPandas, Shapely, STRtree, Folium, pandas, matplotlib, SQL

Tools and Workflows: Excel, SPSS, openLCA, OpenStreetMap, Overpass API, PVGIS, Git, GitHub, Leaflet, AutoCAD

Languages: Arabic (Native), English (C1), Japanese (C1), German (A2-B1)

CERTIFICATES

- ArcGIS API for Python Fundamentals, Esri, 2026
- ArcGIS Pro Scripting and Spatial Analysis Automation, Esri, 2026